

Measuring College STEM Students' Identity, Attitude, and Career Ambition: Sponsored through NSF IUSE

Introduction

Undergraduate research is a high impact practice [1] in which participants have shown benefits with gains in disciplinary knowledge and skills [2, 3], persistence [4, 5], STEM identity and attitudes [6, 7, 8], transferable skills [9, 10], and career ambition [11]. Although being online provides unique challenges for students to participate in research, online students have similar interests in participating in undergraduate research as their residential counterparts [12, 13]. Online faculty also have comparable interest as residential faculty in mentoring undergraduate research [14].

To address the unique challenges of making online research assessable to online students, the Research Scholars Program (RSP) at Embry-Riddle Aeronautical University (ERAU), Worldwide campus, was developed through NSF IUSE #2021221. The RSP was further refined for development, testing, and use of curricular innovations through NSF IUSE #2315560. A major component of the RSP was to build a framework of support [15, 16] that includes a centralized resource center, workshops [17], and research mentoring [18, 19].

It is expected that participating in the RSP will benefit online undergraduate students, specifically in regard to STEM identity, attitude, and career ambition, but a validated process is needed to measure any effects. This paper discusses the process in which surveys validated to measure K-12 students' STEM identity, attitude, and career ambition were applied to college students (both STEM and non-STEM majors). A brief summary of the survey, the validation process, and results are provided. Once validated, these surveys provide a means to show any differences between STEM and non-STEM college students, and they can be used to test the effect of the RSP on participants.

Literature Review

Three psychological constructs have emerged as powerful indicators of a student's engagement with and commitment to their STEM pathway. STEM identity is defined in this study as the overlapping dimensions of one's competence in STEM knowledge, performance of STEM disciplinary practices, and both personal and external recognition of their status as a STEM person (adapted from [20]). While the three-pillar model of competence, performance, and recognition was conceptualized for science identity, STEM can be viewed as a coherent professional identity beyond its individual sub-disciplines [21]. STEM attitude is a student's feelings and beliefs

towards STEM subjects [22]. STEM career ambition reflects the plans and aspirations for the student's future profession [11]. The interplay between these psychosocial factors may provide a self-reinforcing cycle that supports success in the academic STEM pipeline.

STEM Identity

Multiple validated instruments exist for measuring STEM identity, each offering advantages and limitations. The single-item STEM-PIO-1 [23] offers simplicity but lacks the multidimensionality needed to unpack the drivers of student attrition. Scales like SciID [24] and RIS-STEM [25] were validated for secondary and elementary student populations, respectively.

The Dou et al. [21] instrument was selected to operationalize and measure STEM identity because of its strong theoretical alignment with Carlone & Johnson [20] pillars. Comprehensively, this instrument moves beyond internal interest and accounts for all three dimensions of STEM identity at a level appropriate for this student population. The Dou instrument allows for a dynamic and granular understanding of identity-related factors that could influence their persistence.

STEM Attitude

Several scales exist for assessing student attitudes towards STEM, ranging from interpersonal influence models [26] to implicit association tests that explore unconscious bias [27] and belief network analysis [28]. While offering unique insights, they lack disciplinary sensitivity required to address the comparative nature of this study's research questions. To determine if significant differences exist between STEM and non-STEM cohorts, the chosen instrument must provide granular data rather than an aggregate or composite score.

The Guzey, Harwell, and Moore [29] scale uses sub-scales to measure attitudes towards science, mathematics, engineering, and technology independently, allowing nuanced exploration of perspectives towards a specific discipline or a general shift in attitudes towards STEM as a whole. This granularity is important for identifying if possible gaps between STEM and non-STEM cohorts are consistent across disciplines or are driven by specific disciplinary factors. The benefits of the structure of this instrument allowing for correlation of constructs and comparative analysis outweighs the limitations of its original validation demographic and maintains face validity for an adult population.

STEM Career Ambition

Similarly, researchers have validated instruments for measuring STEM ambition, with differences in theoretical approach. Some researchers argue that existing scales conflate social cognitive proxy factors like self-efficacy with the primary factor of career aspiration, arguing this inhibits construct validity [30].

The Carpi et al. [11] instrument was selected as it treats career ambition as a discrete outcome variable. By treating aspiration as a standalone measure, it avoids the conflation issue noted by Fitzgerald et al. [30]. This allows for a statistically straightforward examination of how identity and attitude correlate with career ambition. Additionally, the Carpi scale focuses on professional commitment over general interest, which provides sensitivity for distinguishing between STEM and non-STEM student cohorts.

Methodology

While validated surveys measure K-12 students' science, technology, engineering, and mathematics (STEM) identity [21], attitude [29], and career ambition [11], these surveys have not been validated for college student populations. This study surveyed 304 STEM and 132 non-STEM undergraduate students at a medium-sized private (not-for-profit) university in the United States to validate surveys associated to each of these affective attributes.

Surveys are validated by identifying latent factors (values that cannot be measured directly) by statistically measuring how collections of measurable values (such as survey questions) compare to one another. For a collection of measurable values to be caused by the latent factor, it is expected that these measurable values to appear in consistent ways across multiple users. Once a structure is set for how the measurable values are assumed to be caused by each latent value, a Structure Equation Model (SEM) can be developed that constructs a linear regression for each latent value. The model fit of the data for the specific model can then be evaluated through various statistical measures such as RMSEA, CFI, GFI, and TLI. These measures are commonly used to evaluate educational models (e.g., Chamberlain Jr. & Faulconer [31]). It is important to note that it is cautioned against hard cutoffs for evaluation metrics as model and data characteristics can affect evaluation metrics [32]. For example, the number of factors, when holding all else constant, tends to decrease CFI and TLI [33]. Therefore, the authors are willing to accept evaluation values slightly below target when the model provides a theoretically-sound structure.

As each of the three surveys has been validated on K-12 student populations, there is a theoretical structure to run a SEM. If this SEM meets the evaluation measure targets, the survey can be considered validated for the college population. If the model does not meet the evaluate measure targets, Confirmatory Factor Analysis (CFA) was performed to identify statistically latent factors based on the data set, and these statistically motivated latent factors can be compared to the theorized K-12 latent factors.

All CFA and SEM were performed using the semopy (Structural Equation Modeling Optimization in PYthon) Python package [34]. All code and de-identified data is available in a GitHub repository [35]. Due to space limitations, all SEM graphics will be omitted but can be found on the GitHub repository.

Results

A summary of statistical measures of model fit for each structural model is provided in Table 1. STEM Identity (2-factor structure for 7 questions) and STEM Ambition (2-factor structure for 5 questions) surveys presented the same latent factor structure and were validated via a SEM along all four validation measures. STEM Attitude (7-factor structure for 28 questions) did not present the same latent factor structure as its associated K-12 survey dataset (4-factor structure for 28 questions), but a SEM based on the presentation of the STEM Attitude questions (7 questions with 4 parts each) was validated along all four validation measures. As noted in the Methodology, the evaluation targets are not hard cutoffs and so the slight underperformance on CFI, GFI, and TLI for the STEM Attitude — Presentation SEM is acceptable as there is a practical explanation for its performance.

Table 1: Statistical Measure of Fit for Structural Equation Models

	Root Mean Square Error of Approximation	Comparative Fit Index	Goodness-of- Fit Index	Tucker-Lewis Index
Target	<i>RMSEA</i> < 0.1	<i>CFI</i> > 0.9	<i>GFI</i> > 0.9	<i>TLI</i> > 0.95
STEM Identity — Theoretical	0.090	0.986	0.981	0.977
STEM Attitude — Theoretical	0.129	0.708	0.680	0.677
STEM Attitude — EFA	0.065	0.933	0.898	0.919
STEM Attitude — Presentation	0.080	0.893	0.858	0.877
STEM Ambition — Theoretical	0.032	0.999	0.996	0.997

*Three models were tested for STEM Attitude in particular: theoretical model validated with K-12 students, statistically driven model based on Exploratory Factor Analysis (EFA), and a practical model based on how the questions were presented to students in the survey.

Discussion

The selection of the Dou et al. [21], Guzey et al. [29], and Carpi et al. [11] instruments provides a robust, holistic psychosocial profile of undergraduate students. While these scales arose from distinct research contexts, their integration serves to ensure that identity, attitude, and career ambition are measured as discrete yet related dimensions of a student's academic experience. Using these instruments concurrently without redundancy is an important step in validating a unified measurement model in higher education. A primary challenge in psychosocial research is the burden on research participants and semantic redundancy when multiple scales are used. This study addresses this by selecting non-overlapping instruments to test the feasibility of a unified measurement model.

Furthermore, this framework specifically addresses the recent critique of existing instruments by decoupling career ambition from its secondary proxies. This framework allows for nuanced correlational analysis, facilitating investigation of how specific internal factors like Dou's social recognition or Guzey's disciplinary-specific attitudes correlate with professional ambition as defined by Carpi. Additionally, by testing the Guzey scale in a post-secondary context, this study bridges the gap between K-12 and higher education measurement. Ultimately, the granularity of this approach provides statistical sensitivity for comparing STEM and non-STEM cohorts, allowing the pinpointing of where the psychosocial profiles diverge and offering a focused lens.

Positive SEM results suggest that K-12 surveys targeting STEM students' identity, attitude, and career ambition can be used with college populations. It also presents evidence that the presentation of survey questions plays a role in patterns of student responses that may impact underlying latent factors.

Conclusion

TEXT

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